

Intended Learning Objectives

In this lab you will:

- Design and simulate a cascode amplifier.
- Learn how to generate and use design charts.
- Study the biasing of the cascode amplifier.
- Investigate the gain, the bandwidth, and the GBW of a cascode amplifier.

PART 1: Sizing Chart

1. We can show that the intrinsic gain of a MOSFET is given by

$$|A_v| \approx g_m r_o = \frac{2I_D}{V_{ov}} \times \frac{V_A}{I_D} = \frac{2V_A}{V_{ov}}$$

Interestingly, the gain only depends on λ and V_{ov} . However, to derive this expression we used $g_m = \frac{2I_D}{V_{ov}}$ which is based on the square-law. For a real MOSFET, if we compute V_{ov} and $\frac{2I_D}{g_m}$ they will not be equal. Let's define a new parameter called V-star (V^*) which is calculated from actual simulation data using the formula

$$V^* = \frac{2I_D}{g_m} \leftrightarrow g_m = \frac{2I_D}{V^*}$$

For a square-law device, $V^* = V_{ov}$, however, for a real MOSFET they are not equal. The actual gain is now given by

$$|A_v| \approx \frac{2V_A}{V^*}$$

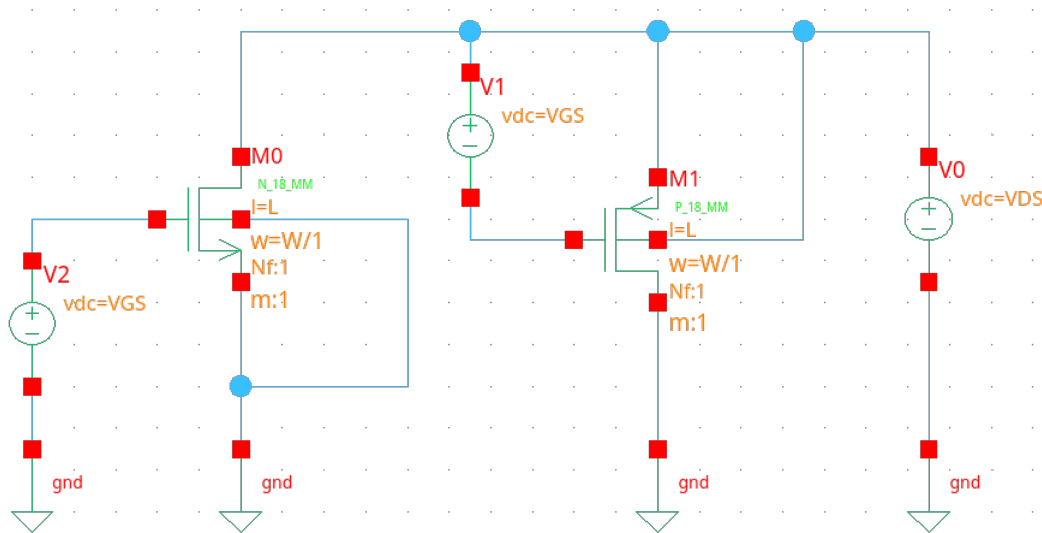
The lower the V^* the higher the gain, but the larger the area and the lower the speed. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200\text{mV}$.

2. We want to design CS and cascode amplifiers with the parameters below.

Parameter	Value
L	0.5 μm
V^*	200 mV
Supply voltage	1.8 V
Current Consumption	20 μA

3. The remaining variable in the design is to calculate W. Since the square-law is not accurate, we cannot use it to determine the sizing. Instead, we will use a sizing chart generated from simulation. Create a testbench for NMOS transistor as shown below (we will use NMOS only in this lab). Use $W = 10\mu\text{m}$ (we will understand why

shortly) and $L=0.5\mu\text{m}$ (the same L selected before). A relatively small L is necessary for Part 3 to make sure that C_{gd} is not negligible compared to C_{gs} .



4. Sweep V_{GS} from 0 to $\approx V_{TH} + 0.4\text{V}$ with 10mV step. Set $V_{DS} = V_{DD}/2$.
5. We want to compare $V^* = 2I_D/g_m$ and $V_{ov} = V_{GS} - V_{TH}$ by plotting them overlaid. Use the calculator to create expressions for V^* and V_{ov} . You can save the expressions to reuse them later.
6. Plot V^* and V_{ov} overlaid vs V_{GS} . Make sure the y-axis of both curves has the same range. You will notice that at the beginning of the strong inversion region, V^* and V_{ov} are relatively close to each other (i.e., square-law is relatively valid). For deep strong inversion (large V_{ov} : velocity saturation and mobility degradation) or weak inversion (near-threshold and sub-threshold operation) the behavior is quite far from the square-law.
7. An often used sweet-spot that provides good compromise between different trade-offs is $V^* = 200\text{mV}$. On the V^* and V_{ov} chart locate the point at which $V^* = 200\text{mV}$. Find the corresponding V_{ovQ} and V_{GSQ} .
8. Plot I_D , g_m , and g_{ds} vs V_{GS} . Find their values at V_{GSQ} . Let's name these values I_{DX} , g_{mX} , and g_{dsX} .
9. Now back to the assumption that we made that $W = 10\mu\text{m}$. This is not the actual value that we will use for our design. But the good news is that I_D is always proportional to W irrespective of the operating region and the model of the MOSFET (regardless square-law is valid or no). Thus, we can use ratio and proportion (cross-multiplication) to determine the correct width at which the current will be $I_{DQ} = 20\mu\text{A}$ as given in the specs. Calculate W as shown below.

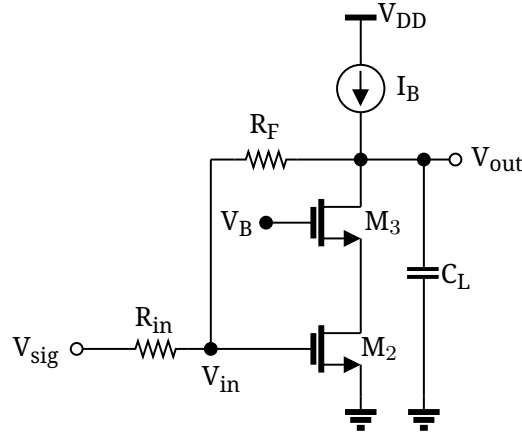
W	I_D
$10\mu\text{m}$	$I_{DX}@V_Q^*$ (from the chart)
?	$I_{DQ}=20\mu\text{A}$ (from the specs)

10. Now we are almost done with the design of the amplifier. Note that g_m is also proportional to W as long as V_{ov} is constant. On the other hand, $r_o = 1/g_{ds}$ is inversely proportional to W (I_D) as long as L is constant. Before leaving this part, calculate g_{mQ} and g_{dsQ} using ratio and proportion (cross-multiplication).

PART 2: Cascode for Gain

1. OP Analysis

- (a) Create a new cell and schematic. Construct the circuit shown below. Use $I_B = 20\mu\text{A}$, $L = 0.5\mu\text{m}$, W as selected in Part 1, and $C_L = 1\text{pF}$.



- (b) Choose V_B (the cascode device bias voltage) such that M_2 has $V_{DS} \approx V^* + 100\text{mV}$ (you may sweep V_B and plot V_{DS} vs V_B to help you choose a good value for V_B).
- (c) We need to bias transistors in saturation; however, the output node is a high impedance node; thus, it is difficult to control its DC voltage. As a workaround in simulation, we use a feedback loop and resistors with different resistances in DC/AC to change the circuit connections in DC/AC simulations (use the AC property in ideal_resistor). The input transistor is diode connected for DC simulation (always in saturation), while in AC simulation the feedback is disconnected, and the AC input source is connected. Set the feedback resistance $1\text{m}\Omega$ DC and $1\text{T}\Omega$ AC and set the source resistance oppositely. We will study how to do biasing practically later in this course.
- (d) Simulate the DC OP point of the above CS and cascode amplifiers. Report a snapshot showing the following parameters for M_1 , M_2 and M_3 in addition to DC node voltages clearly annotated.

Parameter	CSA MOSFET	Cascode Amplifier	
		Input MOSFET	Cascode MOSFET
I_D			
V_{GS}			
V_{DS}			
V_{TH}			
V_{DSAT}			
g_m			
g_{ds}			
g_{mb}			
C_{db}			
C_{gd}			
C_{gs}			
C_{sb}			
Region			

“vdsat” is the minimum drain-source voltage required to bias the transistor in saturation. It is equal to V_{ov} for a square-law device. It is also referred to as “vdss” (drain-source saturation voltage).

- (e) Check that all transistors operate in saturation.
- (f) Do all transistors have the same v_{th} ? Why?
- (g) What is the relation ($\ll$, $<$, $\approx$, $>$, $\gg$) between g_m and g_{ds} ?
- (h) What is the relation ($\ll$, $<$, $\approx$, $>$, $\gg$) between g_m and g_{mb} ?
- (i) What is the relation ($\ll$, $<$, $\approx$, $>$, $\gg$) between c_{gs} and c_{gd} ?
- (j) What is the relation ($\ll$, $<$, $\approx$, $>$, $\gg$) between c_{sb} and c_{db} ?

2. AC Analysis

- (a) Create a new simulation configuration. Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade) to simulate gain and bandwidth.
- (b) Use calculator to create expressions for circuit parameters (DC gain, BW, GBW, and UGF) and export them to adexl.
 - **Cadence Hint:** Use the following expressions in the calculator, and send them to adexl to quickly calculate circuit parameters.

Type	Details	Plot
expr	dB20(VF("/Vout2"))	☒
expr	ymax(dB20(VF("/Vout2")))	☒
expr	ymax(mag(VF("/Vout2")))	☒
expr	bandwidth(VF("/Vout2") 3 "low")	☒
expr	gainBwProd(VF("/Vout2"))	☒
expr	dB20(VF("/Vout1"))	☒
expr	ymax(dB20(VF("/Vout1")))	☒
expr	ymax(mag(VF("/Vout1")))	☒
expr	bandwidth(VF("/Vout1") 3 "low")	☒
expr	gainBwProd(VF("/Vout1"))	☒

- (c) Report the Bode plot (magnitude) of CS and cascode appended on the same plot.
- (d) Using small signal parameters from OP simulation, perform hand analysis to calculate DC gain, BW, and GBW of both circuits.
- (e) Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis.
- (f) Comment on the results.

PART 3 [Optional]: Cascode for BW

1. OP Analysis

- (a) Create a new cell and schematic. Copy the old schematic instances to the new one. Make the following modifications: remove the feedback resistance, set $C_L = 1\text{fF}$, replace the current source with a resistor R_D , make the signal source resistance = $10\text{M}\Omega$ (remove the AC value). Set V_{GS} of the input device as calculated in Part 1.
- (b) Calculate R_D analytically such that the voltage drop on it is $\approx V_{DD}/2$ (the current remains roughly the same as in Part 2 because we are using the same V_{GS}).
- (c) Note that the DC voltage of the output node is set by the resistance (R_D); thus, we don't need a feedback loop as in the previous case.
- (d) Simulate the DC OP point of the new CS and cascode amplifiers. Add monitors and report a snapshot showing the following parameters for M_1 , M_2 and M_3 in addition to DC node voltages clearly annotated.

Parameter	CSA MOSFET	Cascode Amplifier	
		Input MOSFET	Cascode MOSFET
I_D			
V_{GS}			
V_{DS}			
V_{TH}			
V_{DSAT}			
g_m			
g_{ds}			
g_{mb}			
C_{db}			
C_{gd}			
C_{gs}			
C_{sb}			
Region			

(e) Check that all transistors operate in saturation.

2. AC Analysis

- Perform AC analysis (1Hz:10GHz, logarithmic, 10points/decade) to simulate gain and bandwidth.
- Use calculator to create expressions for circuit parameters (DC gain, BW, GBW, and UGF) and export them to adexl.
- Report the Bode plot (magnitude) of CS and cascode appended on the same plot.
- Using small signal parameters from OP simulation, perform hand analysis to calculate DC gain, BW, and GBW of both circuits.
- Report a table comparing the DC gain, BW, UGF, and GBW of both circuits from simulation and hand analysis. Comment on the results.

Lab Summary

- In Part 1 you learned
 - How to generate and use design charts for NMOS and PMOS transistors.
 - How to design common-source and cascode amplifiers.
- In Part 2 you learned
 - How to do ac and DC simulations of a cascode amplifier with current-source load.
 - How the gain of a cascode amplifier with current-source load changes with frequency.
 - How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with current-source load.

- In Part 3 you learned
 - How to do ac and DC simulations of a cascode amplifier with resistive load.
 - How the gain of a cascode amplifier with resistive load changes with frequency.
 - How to simulate the gain, the bandwidth and the GBW of a cascode amplifier with resistive load.